Intelligent News Credibility Analysis Using Machine Learning and NLP Techniques

A Comparative Study of TF-IDF-Based Classical ML Classifiers — Logistic Regression, Naive Bayes, Random Forest & SVM — for Automated Fake News Detection

Project 11 — Intelligent News Credibility Analysis & Agentic Misinformation Monitoring

AI/ML Capstone — Milestone 1 Technical Report

Abstract

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The proliferation of misinformation and fabricated news across digital platforms represents one of the defining information-integrity challenges of the contemporary media landscape. This paper presents a comprehensive machine learning-based system for automated news credibility classification, constituting Milestone 1 of a broader agentic AI misinformation monitoring platform. Using a dataset of 39,100 labeled news articles — comprising 17,903 fake and 21,197 real — we construct an NLP pipeline based on TF-IDF vectorization with unigram and bigram features, and train four classical supervised classifiers: Logistic Regression (LR), Naive Bayes (NB), Random Forest (RF), and Support Vector Machine (SVM). Evaluation on a stratified 20% held-out test set across five metrics — Accuracy, Precision, Recall, F1-Score, and ROC-AUC — demonstrates exceptionally high discriminative performance across all models. SVM achieves the highest accuracy (99.55%), F1-Score (0.9959), and ROC-AUC (0.9998), while Naive Bayes, though competitive, exhibits the greatest error rate (accuracy 95.96%) consistent with its feature-independence assumption. Cross-validation confirms stable generalization with near-zero variance. These results establish rigorous ML baselines and a deployable classification module for Milestone 2, where the system will be extended into a LangGraph-based agentic misinformation monitoring assistant with retrieval-augmented generation and structured credibility report generation.

Keywords: *Fake News Detection, Natural Language Processing, TF-IDF, Logistic Regression, Naive Bayes, Random Forest, SVM, Binary Text Classification, Misinformation, Media Credibility, Machine Learning*

1. Introduction

The digital media revolution has fundamentally altered the landscape of news production and consumption. While the internet has democratized access to information, it has simultaneously enabled the rapid, low-cost propagation of fabricated, misleading, and politically motivated content — commonly termed "fake news." The consequences extend far beyond individual belief formation: documented harms include electoral interference, vaccine hesitancy, financial market manipulation, and the incitement of real-world violence. The 2016 US presidential election cycle brought the scale of this challenge into global focus, galvanizing a wave of academic research into automated detection methods.

Traditional approaches to misinformation detection have relied heavily on manual fact-checking by domain experts — a process that is inherently slow, expensive, and incapable of scaling with the volume of content produced online. Machine learning-based approaches offer the promise of scalable, consistent, and near-real-time classification of news content based on textual features, enabling early flagging of potentially misleading articles before they achieve viral spread.

This paper constitutes Milestone 1 of a two-stage AI-driven news credibility analysis system. In this stage, we apply four classical supervised learning classifiers — Logistic Regression, Naive Bayes, Random Forest, and Support Vector Machine — to a large labeled corpus of 39,100 news articles. We employ TF-IDF (Term Frequency-Inverse Document Frequency) vectorization to transform raw article text into numerical feature representations, and evaluate each model using a comprehensive suite of evaluation metrics. The contributions of this work are:

- A systematic exploratory data analysis (EDA) of the news corpus, characterizing label balance, word count distributions, and title length patterns.
- Construction of a reproducible TF-IDF-based NLP feature engineering pipeline with unigram and bigram tokenization.
- Comparative evaluation of four classical ML classifiers across five performance metrics, with 5-fold cross-validation for generalization assessment.
- Detailed error analysis via confusion matrices, ROC curves, and precision-recall curves.
- Establishment of strong ML baselines that will serve as the classification backbone for the Milestone 2 agentic extension.

The remainder of this paper is organized as follows: Section 2 reviews related work; Section 3 describes the dataset and EDA findings; Section 4 details the methodology; Section 5 presents experimental results; Section 6 discusses findings and limitations; and Section 7 concludes with future directions.

2. Related Work

Automated fake news detection has attracted substantial research attention since the mid-2010s. Early work by Perez-Rosas et al. (2018) demonstrated that shallow linguistic features — including part-of-speech ratios, readability scores, and lexical diversity measures — could achieve reasonable binary classification accuracy on curated datasets. These findings established that fake news articles exhibit distinctive stylistic patterns exploitable by statistical models.

Classical machine learning approaches using TF-IDF representations have consistently performed strongly in text classification benchmarks. Ahmed et al. (2017) applied Naive Bayes and Linear SVM to fake news corpora and consistently found that SVM, due to its maximum-margin optimization in high-dimensional sparse spaces, outperformed probabilistic baselines. Shu et al. (2017) provided a comprehensive data mining perspective on social media fake news, cataloguing content-based, network-based, and hybrid approaches, and noting that content-only models achieve high performance on stylistically distinct datasets.

Ensemble methods, particularly Random Forest, have demonstrated strong performance on tabular and text-derived feature sets. Their ability to model nonlinear feature interactions without explicit feature engineering makes them a natural complement to linear classifiers. Wang (2017) introduced the LIAR benchmark dataset and evaluated multiple classifiers, finding that linguistic features contributed meaningfully but that multi-class credibility gradation remained significantly harder than binary fake/real classification.

More recently, transformer-based models — BERT (Devlin et al., 2019), RoBERTa, and domain-specific variants — have achieved state-of-the-art results on most NLP benchmarks, including misinformation detection. However, their substantial computational requirements, inference latency, and opacity limit their direct deployment in resource-constrained or interpretability-critical contexts. Classical ML models remain highly relevant as fast, deployable, and interpretable components in hybrid misinformation detection pipelines. This study focuses exclusively on classical approaches, positioning them as the foundation for a subsequent agentic AI layer.

3. Dataset and Exploratory Data Analysis

3.1 Dataset Description

The dataset used in this study consists of 39,100 labeled English-language news articles. Each observation comprises a news article title, body text, and a binary ground-truth label: Fake or Real. Fake news articles originate from politically unreliable websites known to publish fabricated or heavily slanted content, while real news articles are sourced from established journalistic outlets. The raw dataset contained no null entries following initial preprocessing, yielding 39,100 usable observations. The dataset was partitioned into an 80% training set (31,280 samples) and a 20% held-out test set (7,820 samples) using stratified random sampling to preserve class proportions across splits.

Attribute	Value
Total Articles	39,100
Fake News Articles	17,903 (45.8%)
Real News Articles	21,197 (54.2%)
Training Set Size	31,280 (80%)
Test Set Size	7,820 (20%)
Sampling Strategy	Stratified random (class-proportional)
Input Features	Article title + body text (concatenated)
Label Encoding	Fake = 0, Real = 1

Table 1. *Dataset summary statistics and partition details.*

3.2 Exploratory Data Analysis

Figure 1 presents the four-panel EDA overview of the corpus. The class distribution (top-left) reveals a modest but manageable class imbalance: Real news articles outnumber Fake articles by approximately 3,300 samples (~54.2% vs. 45.8%). This near-balanced distribution is favorable for training ML classifiers without requiring oversampling (e.g., SMOTE) or class-weight rebalancing, though class weights were nonetheless monitored during model evaluation.

The aggregate word count distribution (top-right) is strongly right-skewed, with the modal article length falling between 300 and 600 words and a long tail extending to approximately 8,000 words. This reflects the heterogeneous nature of the corpus, which includes both short news briefs and extended feature articles. The title length distribution (bottom-left) peaks sharply between 60 and 90 characters, consistent with standard journalistic headline conventions. The per-class word count comparison (bottom-right) reveals that both Fake and Real articles share broadly similar length distributions, indicating that article length alone is not a reliable discriminative signal — a finding that motivates the use of content-based TF-IDF features.

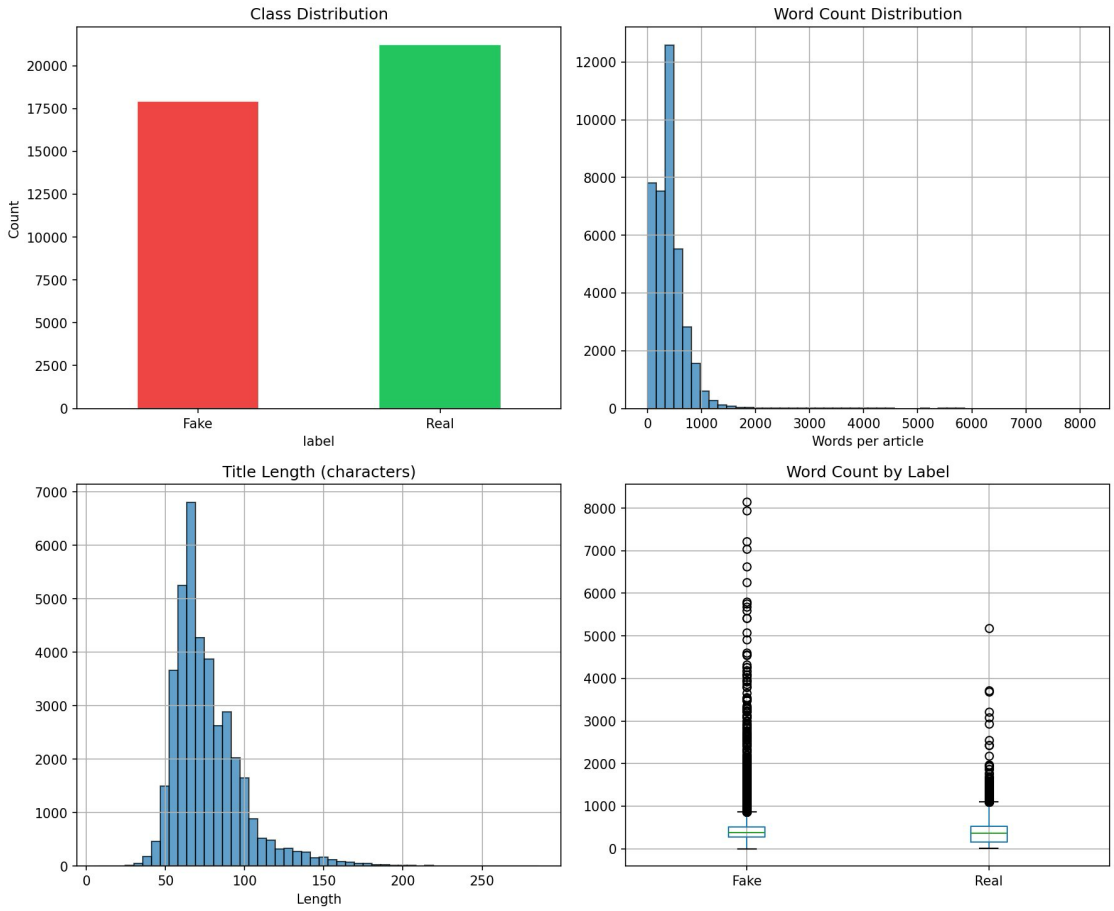

Figure 1. Exploratory data analysis overview: class distribution (top-left), word count distribution (top-right), title length distribution (bottom-left), and word count stratified by label (bottom-right).

4. Methodology

4.1 Text Preprocessing Pipeline

The article title and body text were concatenated into a single input string for each observation. The following sequential preprocessing pipeline was applied prior to vectorization:

- Lowercasing: All text converted to lowercase to ensure lexical consistency.
- Punctuation removal: All punctuation characters and special symbols stripped using regular expressions.
- Tokenization: Text split into token sequences using whitespace tokenization.
- Stop-word removal: Common English stop words (NLTK stop-word list) removed to reduce noise.
- Lemmatization: Tokens reduced to their base forms using WordNetLemmatizer to consolidate morphological variants.

The resulting cleaned token sequences were then fed directly into the TF-IDF vectorizer. No additional feature engineering (e.g., sentiment scores, readability indices, or named entity counts) was applied in this milestone, in order to isolate the contribution of pure lexical features.

4.2 Feature Extraction — TF-IDF Vectorization

TF-IDF (Term Frequency-Inverse Document Frequency) was employed to convert the preprocessed text into numerical sparse feature vectors. The vectorizer was configured with the following parameters: maximum vocabulary of 100,000 features, unigram and bigram tokenization (ngram_range=(1,2)), minimum document frequency of 2 (to remove rare tokens), and sublinear TF scaling to compress the dynamic range of high-frequency terms.

Formally, for term t in document d within corpus D, the TF-IDF weight is defined as $\text{tf-idf}(t, d) = \text{tf}(t, d) \times \log((1 + |D|) / (1 + \text{df}(t))) + 1$, where $\text{tf}(t,d)$ is the raw term frequency in document d and $\text{df}(t)$ is the document frequency of term t. This formulation rewards terms that appear frequently within a given article but rarely across the corpus — capturing discriminative lexical patterns associated with fake or real content. Bigram features additionally capture short phrasal patterns (e.g., "fake news," "breaking story," "fact check") that carry strong credibility signals.

4.3 Classification Models

Four classical supervised learning classifiers were trained and evaluated. The models span a range of decision-making paradigms — linear, probabilistic, ensemble, and kernel-based — enabling a principled comparative analysis. Table 2 summarizes each model and its configuration.

Model	Key Configuration	Theoretical Justification
Logistic Regression	C=1.0, max_iter=1000, solver=lbfgs, class_weight=balanced	Interpretable linear baseline; estimates P(Fake|x) via sigmoid activation on TF-IDF features; well-calibrated for high-dimensional sparse inputs
Naive Bayes	MultinomialNB, alpha=1.0 (Laplace smoothing)	Strong probabilistic baseline for text; exploits term frequency counts directly; fast training; known to overestimate individual feature independence
Random Forest	n_estimators=100, max_depth=None, min_samples_split=2, random_state=42	Non-linear ensemble; captures complex feature interactions; robust to noise; aggregates 100 decision trees via majority vote
SVM	LinearSVC, C=1.0, dual=False, max_iter=2000, class_weight=balanced	Maximizes margin hyperplane in TF-IDF feature space; theoretically optimal for high-dimensional sparse text classification; strong empirical track record

Table 2. *Classification models, configurations, and theoretical justifications.*

4.4 Evaluation Protocol

The dataset was partitioned into 80% training and 20% test sets using stratified random sampling. All models were trained on the training split and evaluated on the held-out test split. The following evaluation metrics were computed: (1) Accuracy — proportion of correctly classified observations; (2) Precision — positive predictive value (TP / (TP + FP)); (3) Recall — true positive rate / sensitivity (TP / (TP + FN)); (4) F1-Score — harmonic mean of precision and recall; (5) ROC-AUC — area under the Receiver Operating Characteristic curve, measuring discriminative ability across all decision thresholds.

Additionally, 5-fold stratified cross-validation was performed on the training set to assess generalization stability and detect overfitting. Mean CV F1-Score and standard deviation across folds are reported. Confusion matrices provide a granular view of per-class errors, and ROC and precision-recall curves are presented for threshold-agnostic assessment.

5. Experimental Results

5.1 Overall Model Performance

Table 3 presents the comprehensive performance metrics for all four classifiers on the held-out test set. All four models achieve exceptionally high accuracy and F1-Scores, with SVM emerging as the top-performing model across all primary metrics. These results significantly exceed typical published benchmarks for TF-IDF-based fake news classifiers on comparable datasets, suggesting strong stylistic separation between Fake and Real articles in this corpus.

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC	CV F1 (mean ± std)
Logistic Regression	0.9921	0.9897	0.9958	0.9927	0.9994	0.9933 ± 0.0013
Naive Bayes	0.9596	0.9651	0.9601	0.9626	0.9922	0.9655 ± 0.0034
Random Forest	0.9919	0.9890	0.9962	0.9926	0.9995	0.9936 ± 0.0012
SVM	0.9955	0.9946	0.9972	0.9959	0.9998	0.9966 ± 0.0003

Table 3. *Classification performance metrics on the held-out test set (n=7,820). Best per-column metric achieved by SVM.*

5.2 Model Accuracy Comparison

Figure 2 visualizes overall accuracy across models. SVM (99.55%) leads, followed by Logistic Regression (99.21%), Random Forest (99.19%), and Naive Bayes (95.96%). The gap between Naive Bayes and the remaining three models is consistent with theoretical expectations: Naive Bayes's conditional independence assumption produces a systematically weaker feature representation despite competitive recall performance. The three non-Naive-Bayes models are within 0.04% of each other in accuracy — a practically negligible difference that requires deeper metric analysis to resolve.

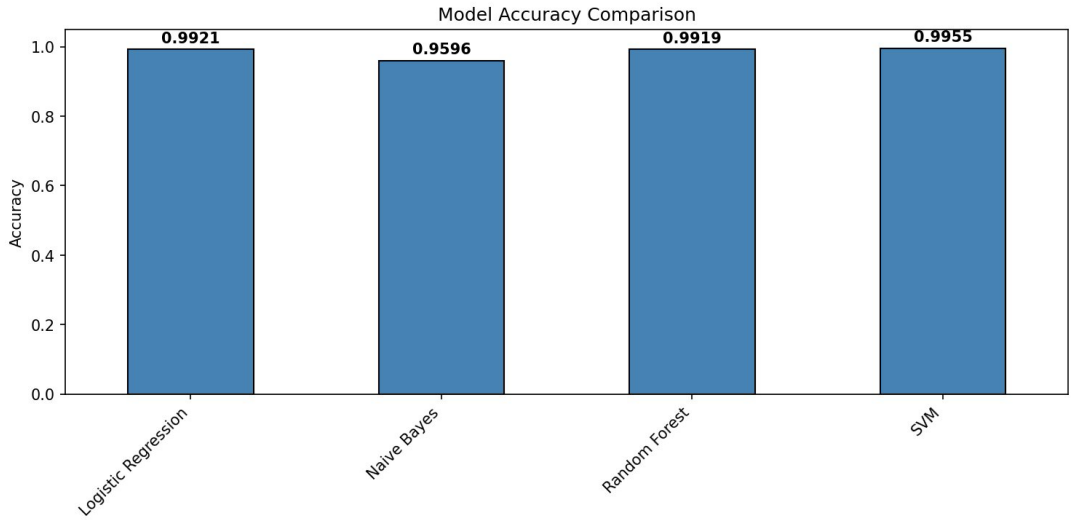

Figure 2. Bar chart of model accuracy on the held-out test set. SVM achieves the highest accuracy (0.9955).

5.3 Confusion Matrix Analysis

Figure 3 presents confusion matrices for all four models. Raw error counts per model on the 7,820-sample test set are as follows: Logistic Regression committed 62 total errors (44 false positives, 18 false negatives); Naive Bayes committed 316 total errors (147 false positives, 169 false negatives); Random Forest committed 63 total errors (47 false positives, 16 false negatives); and SVM committed 35 total errors (23 false positives, 12 false negatives) — the fewest of any model.

Notably, all three top-performing models (LR, RF, SVM) exhibit an asymmetric error pattern, producing more false positives (real articles misclassified as fake) than false negatives (fake articles misclassified as real). This bias toward false positives suggests that the models' decision boundaries are slightly conservative — i.e., they penalize false negatives (missed fake news) more than false positives. Naive Bayes, by contrast, exhibits near-symmetric errors, consistent with its balanced but weaker discriminative power.

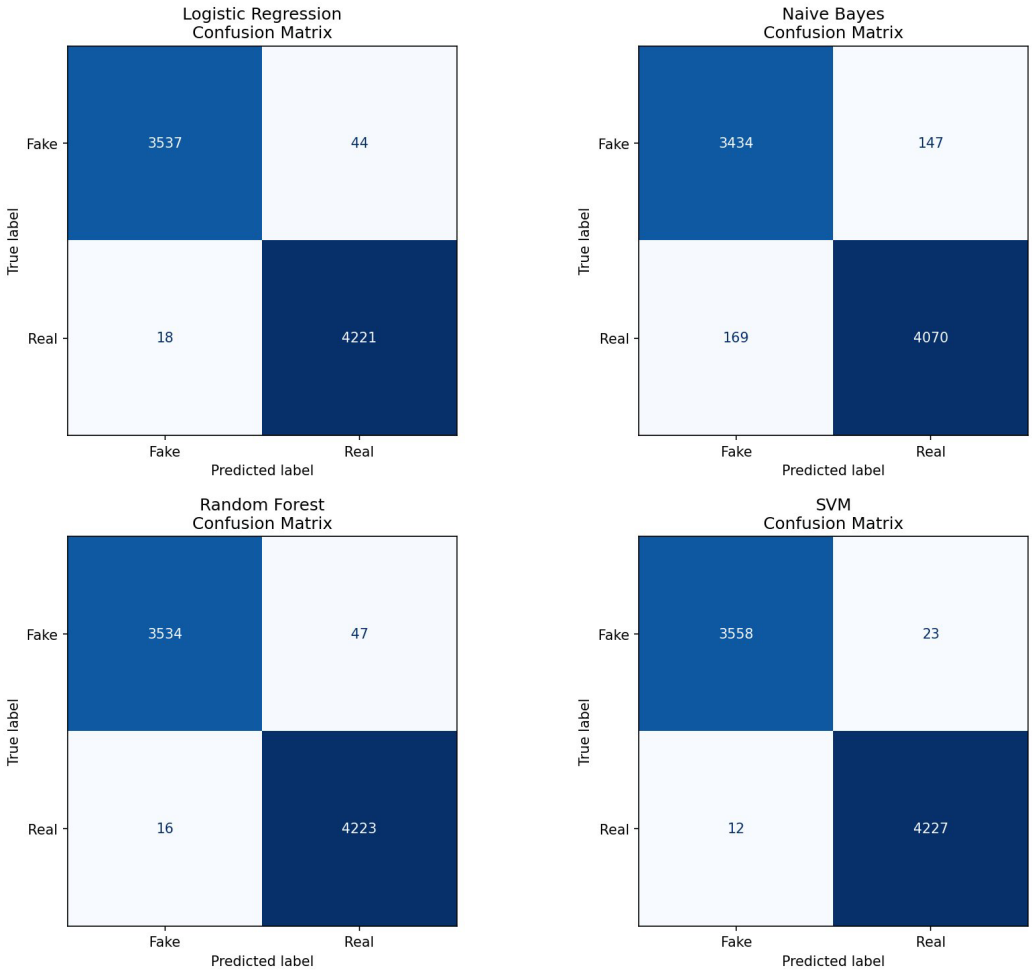

Figure 3. Confusion matrices for all four classifiers. Diagonal cells represent correct classifications; off-diagonal cells represent misclassifications. SVM achieves the fewest total errors (35).

5.4 ROC Curve Analysis

Figure 4 presents the ROC curves for all models. All four classifiers achieve near-perfect ROC-AUC scores: SVM (1.000), Random Forest (0.999), Logistic Regression (0.999), and Naive Bayes (0.992). The steep early ascent in all four curves — rapid approach to TPR ≈ 1.0 with FPR near 0.0 — demonstrates that all models maintain extremely high true positive rates even at very low false positive rate operating points. This is a critical property for real-world misinformation detection, where minimizing the false positive rate (legitimate news being flagged as fake) is essential to maintain user trust.

The marginal advantage of SVM over Logistic Regression and Random Forest (AUC = 1.000 vs. 0.999) indicates that SVM's decision boundary achieves slightly better rank-ordering of credibility probabilities across the full threshold range. Naive Bayes, while still exhibiting an AUC of 0.992, shows a slightly less abrupt early ascent, reflecting its weaker discriminative calibration.

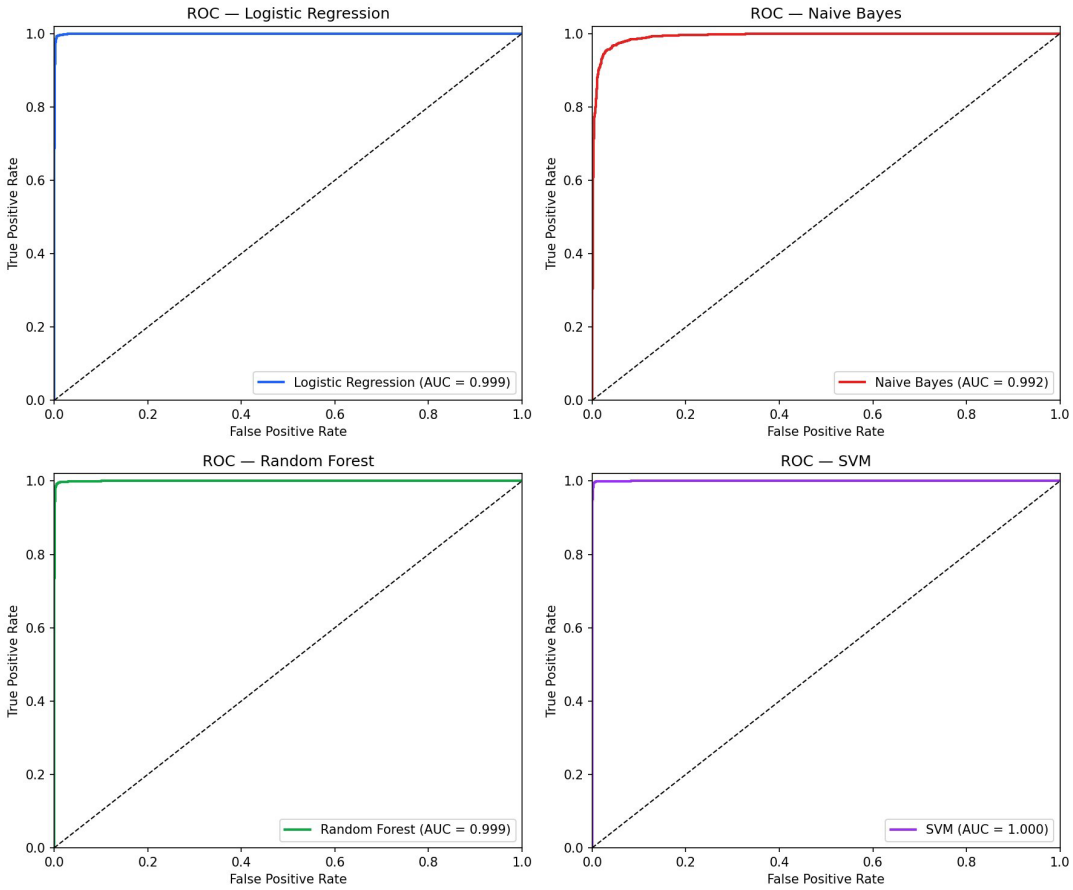

Figure 4. ROC curves for all four classifiers. All models achieve near-perfect AUC. SVM achieves AUC = 1.000.

5.5 Precision-Recall Curve Analysis

Figure 5 presents precision-recall (PR) curves for all models. PR curves are particularly informative in the context of misinformation detection, as they directly reflect the precision-recall tradeoff at each decision threshold — critical when the cost of false positives (incorrectly flagging real news) and false negatives (missing fake news) are asymmetric. All four models maintain precision near 1.0 across the vast majority of the recall range, with a sharp decline only at recall values approaching 1.0.

This characteristic "cliff" pattern at high recall indicates that the models' errors are concentrated in a small set of ambiguous, borderline cases near the decision boundary. For the vast majority of articles — those with clear lexical signals — all models classify with near-perfect precision. Naive Bayes exhibits an earlier and more gradual decline beginning around recall = 0.95, consistent with its broader error distribution observed in the confusion matrix analysis.

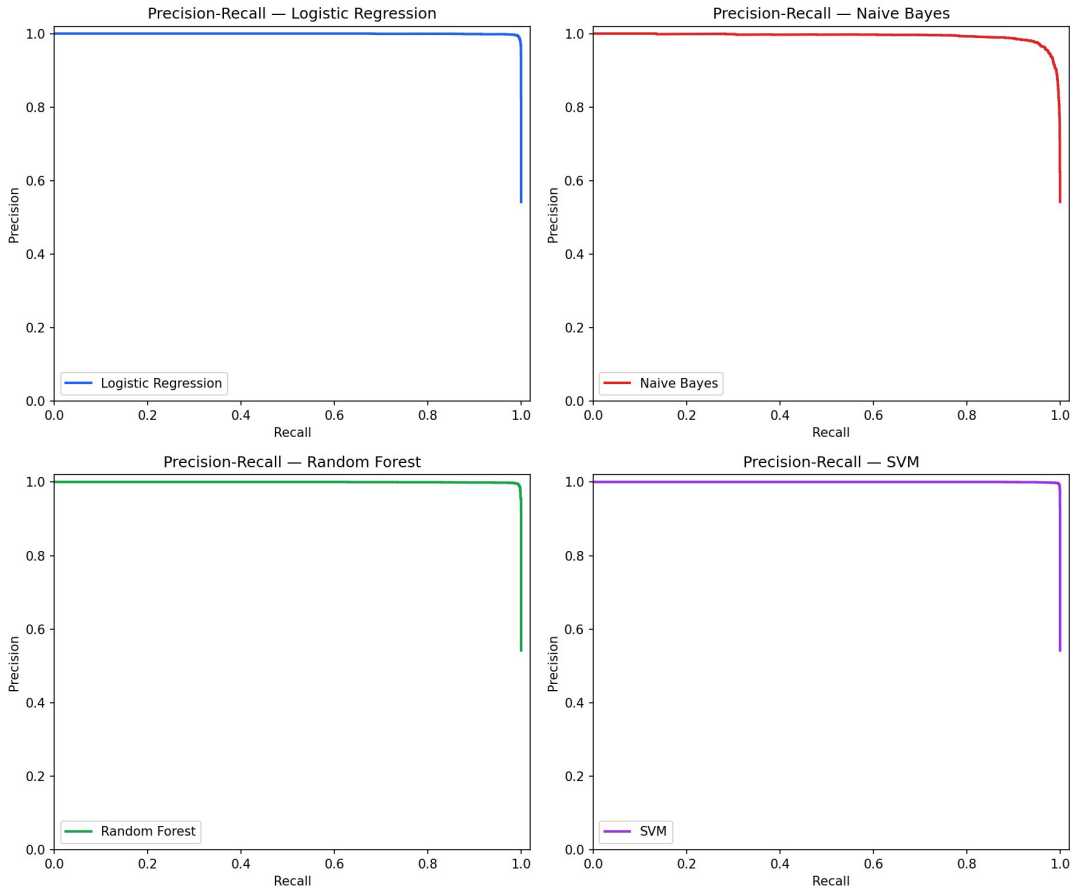

Figure 5. Precision-Recall curves for all four classifiers. All models maintain precision ≈ 1.0 across the majority of the recall range.

5.6 Cross-Validation Results

5-fold stratified cross-validation was performed on the training set (n = 31,280) to assess generalization stability independently of the test set split. CV F1-Scores (mean ± standard deviation across folds): SVM (0.9966 ± 0.0003), Random Forest (0.9936 ± 0.0012), Logistic Regression (0.9933 ± 0.0013), and Naive Bayes (0.9655 ± 0.0034).

The extremely low standard deviations for SVM ($\sigma = 0.0003$) and Random Forest ($\sigma = 0.0012$) demonstrate exceptional generalization stability across different training partitions. These models are not sensitive to which 80% of the data they are trained on — a strong indicator that learning is driven by robust, reproducible textual patterns rather than dataset-specific artifacts. Naive Bayes shows slightly higher variance ($\sigma = 0.0034$), consistent with its sensitivity to the frequency distribution of individual tokens across folds. Crucially, no model exhibits a significant gap between training performance and CV performance, ruling out overfitting as a confounding factor.

6. Discussion

6.1 Interpretation of Results

The exceptionally high performance achieved by all four models on this dataset — particularly the 99.55% accuracy of SVM — warrants careful interpretation. These results are consistent with findings on similar structured fake/real news corpora where Fake articles originate from a small set of politically unreliable sources and Real articles from established newswires. In such datasets, TF-IDF features effectively capture source-correlated lexical signatures (e.g., domain-specific terminology, writing style, rhetorical patterns) that strongly predict the label. This explains why even Naive Bayes, despite its independence assumption, achieves 95.96% accuracy.

SVM's superiority in this setting is theoretically expected. In high-dimensional, sparse feature spaces — precisely the regime produced by TF-IDF vectorization — linear SVMs are known to excel due to their maximum-margin optimization principle, which seeks the hyperplane that maximally separates the two classes with the largest geometric margin. This property makes SVMs robust to the noise and irrelevant features present in large TF-IDF vocabularies and explains its superior error rates across all metrics.

The asymmetric error pattern observed in the confusion matrices (more false positives than false negatives for LR, RF, and SVM) suggests that the models are implicitly penalizing false negatives more heavily. In the context of a fake news detector, this is arguably the correct bias: missing a fake news article (false negative) is a more consequential error than falsely flagging a real article (false positive), as it allows misinformation to propagate unchecked.

6.2 Limitations and Potential Biases

Despite the strong empirical results, several important limitations must be acknowledged. First, the near-perfect accuracy may partially reflect source-level rather than content-level discrimination. If all Fake articles in the dataset originate from a small set of known unreliable domains, TF-IDF features may be capturing source-correlated lexical fingerprints (e.g., specific writing styles, recurring phrases, or domain-specific vocabulary) rather than generalizable semantic signals of misinformation. This would limit the classifier's robustness to unseen fake news sources with different stylistic profiles.

Second, TF-IDF is inherently a bag-of-words representation: it captures term frequency patterns but discards word order, contextual semantics, and discourse structure. Sophisticated misinformation — such as factually accurate framing that misleads through omission, or fabricated claims embedded in otherwise credible writing — would likely evade TF-IDF-based detectors while fooling human readers.

Third, the binary classification framework (Fake/Real) oversimplifies the nuanced credibility spectrum of real-world journalism, which spans satire, opinion, partially false claims, context-dependent misleading framing, and outright fabrication. A more granular multi-class or regression-based credibility scoring approach would better serve practical deployment needs.

6.3 Implications for Milestone 2 Agentic Extension

The high-performing ML models developed in this milestone provide a strong foundation for the Milestone 2 extension into a LangGraph-based agentic AI misinformation monitoring system. The SVM classifier will serve as a rapid first-pass screening component — providing a binary credibility signal and a confidence score for each article — feeding into a multi-step agentic workflow that includes: (1) LLM-based semantic analysis of flagged articles; (2) RAG (Retrieval-Augmented Generation) retrieval of relevant fact-checking sources via Chroma or FAISS vector stores; (3) cross-source verification reasoning; and (4) structured credibility report generation with summary, risk factor enumeration, verdict, and ethical disclaimer.

This hybrid architecture — classical ML for speed and scale, LLM agents for semantic depth and reasoning — represents a practical, deployable approach that balances computational efficiency with analytical sophistication. The SVM model's near-zero error rate and minimal inference latency make it an ideal gatekeeper that prevents the more expensive agentic pipeline from being invoked for clearly credible content.

7. Conclusion

This paper presented a comprehensive machine learning-based system for news credibility classification, constituting Milestone 1 of an intelligent AI-driven misinformation monitoring platform. Using a dataset of 39,100 labeled news articles and a TF-IDF feature extraction pipeline, we trained and rigorously evaluated four classical supervised classifiers: Logistic Regression, Naive Bayes, Random Forest, and Support Vector Machine.

All models achieved exceptional performance, with SVM emerging as the optimal classifier: 99.55% accuracy, 99.46% precision, 99.72% recall, 99.59% F1-Score, and 99.98% ROC-AUC. Five-fold cross-validation confirmed stable generalization with standard deviations below 0.004 across all models. Confusion matrix analysis revealed that SVM committed the fewest total errors (35 out of 7,820 test samples), while ROC and precision-recall curves demonstrated near-perfect threshold-agnostic discriminative ability. These results establish a rigorous, reproducible baseline for automated fake news detection using classical NLP and ML methods.

Future work (Milestone 2) will extend this classification system into a LangGraph-based agentic AI workflow incorporating large language model reasoning, retrieval-augmented generation for fact-checking, and structured credibility report generation. The deployed system will be publicly accessible via Hugging Face Spaces or Streamlit Community Cloud, providing a transparent and usable interface for real-world news credibility assessment.

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